DIPLOMA ELECTRONICS / ECE PROGRAM ELECTIVE

Medical Electronics

Full-screen handbook for 6041A. It is not a simple notebook. It covers biomedical signal recording, analytical equipment, therapeutic equipment, imaging systems, biotelemetry and medical electrical safety with diagrams, revision and model paper.

COURSE CODE
6041A

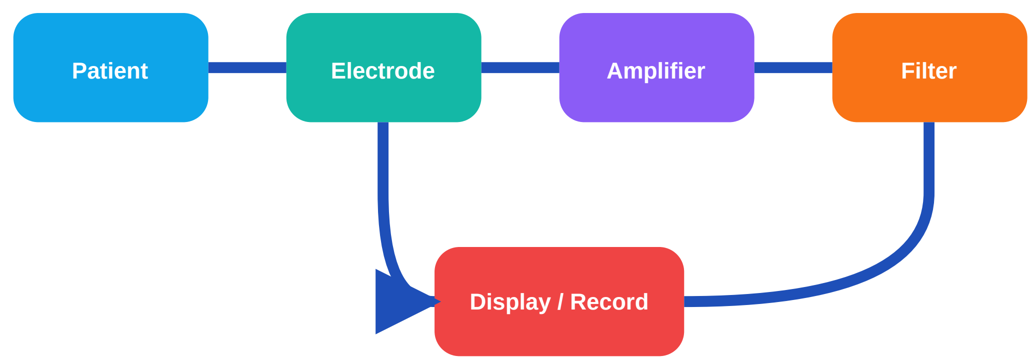
SEMESTER
6

CREDITS
4

PERIODS
60

Start handbook **Model paper**

ECG, EEG, EMG, BP, blood gas analyser, pulse oximeter, LASER, pacemaker, defibrillator, dialysis, ventilator, diathermy, X-ray, CT, MRI, ultrasound, biotelemetry, macroshock, microshock □□□□□□□ exam answer style-□ □□□□□□□□□.



Biomedical instrumentation chain

Official Syllabus Structure

Course objectives

- Familiarize electronics applications in biomedical field.
- Explain analytical equipment, therapeutic equipment, biotelemetry and imaging systems.
- Describe safety considerations in medical field.

Prerequisites

Transducers, actuators, amplifiers and oscillators.

CO	Outcome	Hours	Focus
CO 1	Illustrate ECG, EEG and EMG.	14	biopotentials, electrodes, ECG waveform, recorders
CO 2	Describe blood parameter measurements and LASER applications.	15	blood cells, BP, blood gases, pulse oximeter, laser
CO 3	Summarize therapeutic equipment.	15	pacemaker, defibrillator, dialysis, ventilators, diathermy
CO 4	Explain imaging systems and biomedical equipment precautions.	14	X-ray, CT, MRI, ultrasound, telemetry, safety

CO1 - 14 HOURS

ECG, EEG and EMG

Biomedical signal is very small. Electrode contact, high input impedance amplifier, filtering and isolation decide waveform quality.

Exam answer pattern: definition, neat block diagram, working flow, applications and safety points.

Handbook explanation

Cardiovascular system

Heart electrical activity produces ECG. SA node, AV node and ventricular conduction create the waveform used for monitoring.

Respiratory and nervous systems

Respiration links to oxygen and carbon dioxide measurement. Nervous activity gives EEG and nerve-muscle activity gives EMG.

Biopotentials

Resting potential exists at inactive cell membrane. Action potential is rapid voltage change during excitation.

Electrodes

Surface electrodes are used for ECG, needle electrodes for detailed EMG and microelectrodes for small biological regions.

ECG waveform

P wave means atrial depolarization, QRS means ventricular depolarization and T wave means ventricular repolarization.

Lead systems

Multiple ECG leads view heart electrical activity from different directions. One lead is not enough for full interpretation.

ECG recorder

Electrodes, lead selector, protection, differential amplifier, isolation, filters, display and recorder are the main blocks.

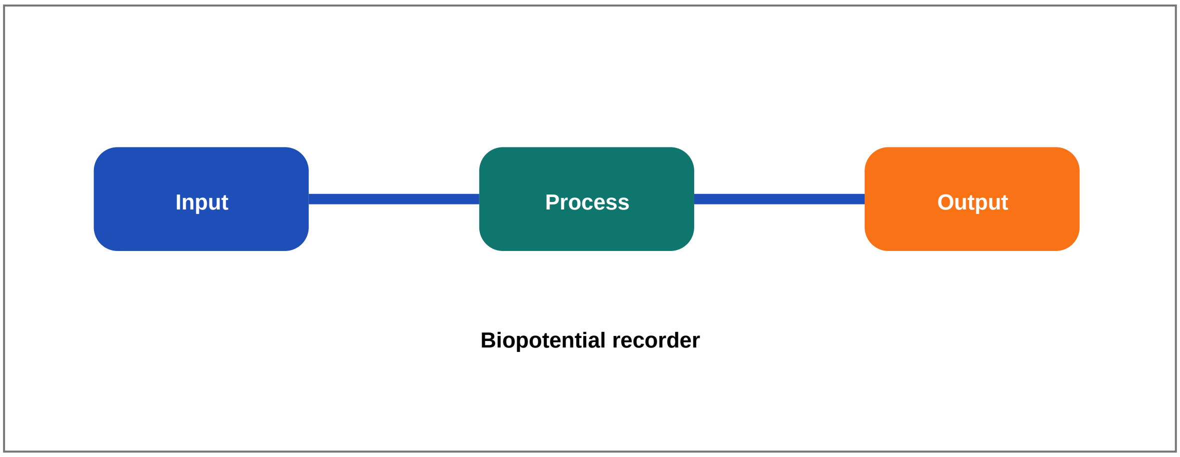
EEG machine

Scalp electrodes, montage selector, low-noise amplifier, filters and recorder measure brain activity.

EMG recorder

Surface or needle electrodes, amplifier, filter and recorder measure skeletal muscle activity.

Diagram



CO2 - 15 HOURS

Blood Measurement and LASER

Blood instruments convert biological parameters into optical, impedance, pressure or electrochemical signals.

Exam answer pattern: definition, neat block diagram, working flow, applications and safety points.

Handbook explanation

Blood cells

RBC, WBC and platelets are counted and classified in hematology instruments.

Microscopic counting

Diluted blood sample is counted in a chamber under microscope. Simple but slow and operator dependent.

Optical counting

Cells interrupt or scatter light and produce pulses for automated counting.

Coulter counter

Cell passing through aperture changes impedance. Pulse count gives number and pulse amplitude relates to size.

Blood pressure

Systolic is maximum pressure, diastolic is minimum pressure and pulse pressure is their difference.

Direct BP

Catheter and pressure transducer give continuous accurate BP but it is invasive.

Indirect BP

Cuff based measurement uses Korotkoff sounds or oscillometric method. It is non-invasive.

Blood gas analyser

Measures PO2, PCO2 and pH using suitable electrodes and membranes.

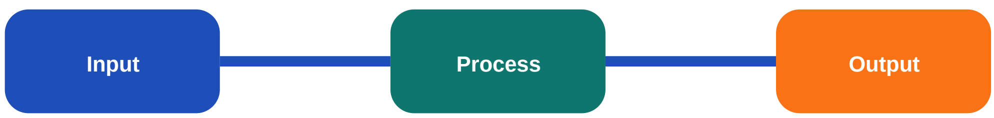
Pulse oximeter

Uses red and infrared light absorption to estimate oxygen saturation non-invasively.

LASER

Stimulated emission produces coherent, monochromatic and directional light used in surgery, ophthalmology and dermatology.

Diagram



Blood sample to result

CO3 - 15 HOURS

Therapeutic Equipment

Therapeutic equipment gives treatment, so working principle plus patient safety is compulsory.

Exam answer pattern: definition, neat block diagram, working flow, applications and safety points.

Handbook explanation

Pacemaker

Artificial pulse source that maintains heart rhythm when natural pacing is inadequate.

Pacemaker modes

External/internal and asynchronous/demand modes. Demand mode senses natural beat before pacing.

Defibrillator

Delivers controlled shock to restore organized rhythm during dangerous arrhythmia.

Hemodialysis

Removes waste through semipermeable membrane using blood and dialysate circuits.

Dialysis safety

Blood pump, pressure monitor, air bubble detector, conductivity and temperature alarms are critical.

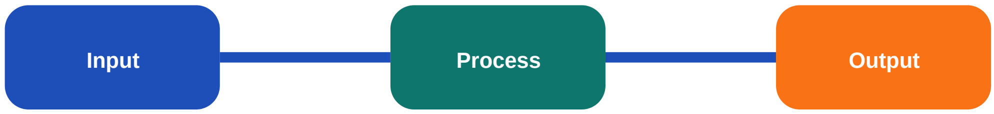
Ventilators

Pressure cycling stops inspiration at set pressure. Volume cycling delivers selected volume.

Diathermy

Shortwave and microwave diathermy produce deep tissue heating using high-frequency energy.

Diagram



Therapeutic chain

CO4 - 14 HOURS

Imaging, Biotelemetry and Safety

Imaging answer order: source, patient interaction, detector, processor, image and safety.

Exam answer pattern: definition, neat block diagram, working flow, applications and safety points.

Handbook explanation

X-ray machine

High voltage tube produces X-rays. Blocks are generator, tube, collimator, patient table, detector and console.

CT scanner

Many X-ray projections are reconstructed by computer to produce cross-sectional images.

Ultrasound

Piezoelectric transducer sends sound pulses and receives echoes. Echo time gives depth.

Echocardiography

Ultrasound imaging of heart showing chamber and valve motion.

MRI

Strong magnetic field and RF pulses produce tissue signals. Magnetic safety is critical.

Biotelemetry

Sensor, amplifier, transmitter, channel, receiver, demodulator and display transmit biosignals remotely.

Macroshock and microshock

Macroshock is external body current. Microshock is small current reaching heart through internal path.

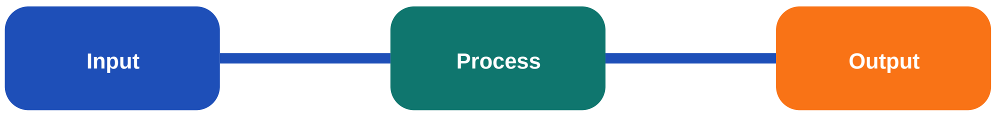
Safety measures

Protective earth, isolation, leakage current control, intact patient leads, calibration, alarms and records are mandatory.

X-ray precautions

Use shielding, collimation, minimum exposure, distance, warning lights and controlled area.

Diagram



Imaging chain

Formula Bank and Revision

ECG: P = atrial depolarization, QRS = ventricular depolarization, T = ventricular repolarization.

Pulse pressure = systolic pressure - diastolic pressure.

MAP approximately equals DBP plus one third of pulse pressure.

Coulter counter: one cell through aperture gives one impedance pulse.

LASER properties: coherence, monochromaticity, directionality and high intensity.

Ultrasound depth is proportional to echo return time.

Macroshock is external body current; microshock is current reaching heart internally.

Expected Questions

Q1 Define ECG, EEG and EMG.

Q2 Explain electrode types.

Q3 Draw ECG recorder block diagram.

Q4 Explain EEG machine.

Q5 Explain EMG recorder.

Q6 Compare cell counting methods.

Q7 Explain BP measurement.

Q8 Explain blood gas analyser.

Q9 Explain pulse oximeter.

Q10 Explain LASER applications.

Q11 Explain pacemaker.

Q12 Explain defibrillator.

Q13 Explain hemodialysis.

Q14 Explain ventilators.

Q15 Explain diathermy.

Q16 Explain X-ray and CT.

Q17 Explain MRI and ultrasound.

Q18 Draw biotelemetry block diagram.

Q19 Explain macroshock and microshock.

Q20 List biomedical safety precautions.

Model Question Paper

Course: 6041A Medical Electronics
Time: 3 Hours
Maximum Marks: 100

Part A
1. Define biopotential and action potential.

2. List electrode types used for ECG, EEG and EMG.
3. State the need of ECG lead system.
4. What is pulse oximetry?
5. State two LASER properties.
6. Define microshock.

Part B

7. Draw and explain ECG recorder block diagram.
8. Explain EEG and EMG measurement with block diagram.
9. Compare microscopic, optical and Coulter counter blood cell counting methods.
10. Explain direct and indirect BP measurement.
11. Explain blood gas analyser and pulse oximeter.
12. Explain pacemaker need, modes and types.
13. Explain defibrillator and hemodialysis machine.
14. Explain ventilator and diathermy equipment.
15. Explain X-ray, CT, MRI and ultrasonic imaging.
16. Explain biotelemetry and medical electrical safety precautions.

Answer Key

1

Biopotential is voltage generated by biological activity. Action potential is rapid excitation voltage change.

2

Surface, needle and micro electrodes are used depending on signal and location.

3

ECG recorder: electrodes, lead selector, protection, amplifier, isolation, filters, display and recorder.

4

EEG uses scalp electrodes, montage selector, low noise amplifier, filters and recorder.

5

EMG uses surface or needle electrode to record muscle electrical activity.

6

Microscopic is manual, optical uses light pulses, Coulter uses impedance pulses.

7

Direct BP uses catheter transducer; indirect BP uses cuff and Korotkoff or oscillometric method.

8

Blood gas analyser measures PO2, PCO2 and pH using electrochemical electrodes.

9

Pulse oximeter uses red and infrared absorption to estimate oxygen saturation.

10

LASER uses stimulated emission and is used in surgery, eye treatment and dermatology.

11

Pacemaker gives artificial pulses; modes include asynchronous and demand.

12

Defibrillator delivers controlled shock to restore organized heart rhythm.

13

Hemodialysis uses semipermeable membrane to remove waste from blood.

14

Ventilators support breathing; pressure and volume cycling are key types.

15

Diathermy produces deep heating using high-frequency energy.

16

X-ray uses high voltage tube; CT reconstructs slices from many projections.

17

MRI uses magnetic field and RF; ultrasound uses echo from sound waves.

18

Telemetry transmits biosignal from sensor to remote receiver and display.

19

Macroshock flows through body; microshock reaches heart internally and is dangerous.

20

Use grounding, isolation, leakage current test, shielding, calibration and alarms.

References

- Biomedical Instrumentation and Measurements - Leslie Cromwell, Fred J Weibell, Erich A Pfeiffer.
- Handbook of Biomedical Instrumentation - R S Khandpur.
- Principles of Medical Electronics and Biomedical Instrumentation - C Raja Rao and S K Guha.
- Medical Instrumentation - John G Webster.

Safety note

This is an engineering study handbook, not a clinical manual. Real patient equipment must follow hospital protocol, manufacturer manual and relevant safety standards.